

Benchmark design governs reported skill in daily river water-temperature prediction

[AUTHOR LIST TO BE COMPLETED]¹

¹[AFFILIATION 1 TO BE COMPLETED — department or laboratory, institution, city, postcode,
country]

²[AFFILIATION 2 TO BE COMPLETED — add or delete affiliation lines to match the final author list]

Key points

- Confirmatory: seven-day skill of +0.250 against persistence falls to +0.038 against damped persistence on identical keys.
- Descriptive: withholding a gauge's own recent water temperature from the inputs costs 1.3-1.7 °C; the model class costs under 0.01 °C.
- Descriptive: future weather is worth 0.13-0.63 °C, and worth less once the local gauge is gone, so the two are complements.

Corresponding author: [CORRESPONDING AUTHOR TO BE COMPLETED],
[INSTITUTIONAL E-MAIL TO BE COMPLETED]

Abstract

Daily river water temperature is strongly persistent and seasonal, so reported machine-learning skill depends on what a model is compared against and on what it may see. On one frozen panel of 120 U.S. gauges across 15 hydrologic regions we vary the evaluation design and the information supplied, scoring every arm on one common registry of station/date/horizon keys with all preprocessing fitted strictly backwards in time. On a pre-specified 2021-2023 held-out window, a constrained deep predictor's seven-day station-median skill is +0.250 against persistence but +0.038 against damped persistence, and a gradient-boosted tree with site identity is most accurate at one and three days. All five sealed tests are supported at their registered margins, two of them non-inferiority tests against a pre-declared +0.05 °C ceiling that carries no ecological interpretation. Descriptive analyses on the same keys measure what the information is worth. Withholding a gauge's own recent water temperature from the inputs costs 1.3-1.7 °C at every lead and at all 116 stations, two orders of magnitude more than the gap between the two estimator implementations compared. Realized future meteorology is worth 0.13-0.63 °C, is almost entirely future air temperature, and survives two pre-registered placebos. Future weather and local observations are complements rather than substitutes: forcing is worth less once those observations are withheld, and the estimator begins to matter only where information is scarce. The held-out comparison is confirmatory; every information result is post-outcome and descriptive, each an input ablation at gauged sites, not prediction where no record exists.

Plain Language Summary

River temperature changes slowly from day to day, so a forecast that repeats yesterday's reading is already fairly accurate, and nudging it toward the season's usual value is better still. We tested predictions at 120 U.S. gauges under one demanding evaluation: every model scored on exactly the same days and sites, none allowed to see anything dated after its issue time. What a model was given mattered far more than which method it used. Denying a model the recent readings from the gauge it is predicting cost roughly two hundred times more accuracy than switching between entirely different families of model. Letting it see the weather that has not happened yet was worth about a third as much, came almost entirely from air temperature, and — unexpectedly — was worth less, not more, once those readings were gone. These are comparisons between what a model is fed,

at rivers that do have thermometers; none of them tests where a thermometer should be installed. Much of what is usually reported as a modelling advance is, on this evidence, a statement about what the model was given and what it was compared against.

Keywords: river water temperature; benchmark design; strong baselines; spatial holdout; comparative evaluation; information value.

1 Introduction

Stream and river thermal regimes set dissolved-oxygen saturation, metabolic rates, life-stage timing, and habitat suitability, and they respond to atmospheric forcing, discharge, groundwater exchange, riparian shading, channel geometry, and regulation (Caissie, 2006). Water managers need daily-resolution temperature at gauged reaches, and the supply of daily records from national monitoring networks has made the variable one of the most heavily modeled targets in applied hydrological machine learning.

The prevailing understanding in that literature is that deep sequence models have substantially advanced daily river-temperature prediction. Nonlinear air–water regressions (Mohseni et al., 1998) and hybrid air-temperature and discharge formulations (Toffolon & Piccolroaz, 2015) were succeeded by recurrent, multi-task, and physics-guided river-network architectures that report large error reductions (Rahmani et al. (2021); Sadler et al. (2022); Barclay et al. (2023)), and graph architectures make spatial dependencies explicit (Topp et al., 2023). Transfer to unseen conditions is a question of spatial sensitivity rather than of raw accuracy, and it is the one this study isolates by holding out whole regions rather than neighbouring sites.

Those reported gains are extraordinarily heterogeneous, and the heterogeneity does not organize itself by architecture. Published accuracies are not comparable across studies because reference models, key sets, and data-splitting conventions differ (Corona & Hogue, 2025). Where studies do compare families under a common protocol, the spread between them is small: neither of two systematic comparisons — one across statistical and machine-learning water-temperature models including an air2stream formulation (Feigl et al., 2021), one across recurrent, convolutional

and transformer architectures for hydrological prediction generally (Liu et al., 2025) — finds an architectural ordering that would account for the published dispersion. The dispersion survives within single panels: on the data used here, one fixed model and one fixed set of prediction days produce a seven-day skill of +0.251 or of +0.038 depending only on which reference model is placed in the denominator. An architectural account cannot produce that spread, because the architecture does not change between the two numbers.

Hydrology has known for a long time that a reported score is a statement about its benchmark, from the differential split-sample test (Kleměš, 1986) through explicit upper and lower benchmarks (Seibert et al., 2018) to benchmarking as the centre of what machine learning can tell us about a hydrological system (Nearing et al., 2021). None of this is contested. What is missing for daily river temperature is not the argument but the measurement: how large the benchmark effect actually is for this variable, on identical prediction keys, with an estimand that pairs models station by station rather than comparing marginal aggregates.

Three design choices make that measurement attributable. First, skill is usually quoted against naive persistence, climatology, or an air–water regression. Daily mean water temperature has large thermal inertia, so relaxing the last observation toward a seasonal climatology already removes most of the error a learned model can remove. Second, models are scored on model-specific complete-case sets, so a model can gain apparent accuracy by declining to predict on hard days, and preprocessing statistics are commonly fitted on periods that include the evaluated interval (Arsenault et al., 2018). Third, spatial generalization is reported from random held-site splits; when a held-out gauge's neighbours remain in training, the quantity measured is closer to interpolation than to transfer (Kratzert et al., 2019). None of these choices is detectable from a reported RMSE, and each can inflate it.

Two boundaries of this study follow from that framing, and recent work sits on the far side of both. Every model here consumes issue-time information only and is scored on retrospective gridded meteorology, which is not what a forecaster could have run: operational stream-temperature forecasting with archived as-issued weather is a different and harder measurement (Zwart, Oliver, et al. (2023); Padrón et al. (2025)). And every arm here retains the target gauge's own observed history,

so nothing in this paper speaks to prediction at reaches that have never been instrumented, which is now being addressed at CONUS scale by frameworks built for exactly that case (Bakar Siddik et al. (2026); Philippus et al. (2026)) and in networks far sparser than this cohort (Chang et al., 2025). We measure how much a benchmark's construction moves a reported number; we do not measure either of those things.

Here we hold the model and the data fixed and vary the design choices instead. We assembled a panel of 657,480 site-days from 120 stable U.S. Geological Survey site numbers spanning 2006–2020, 34 states, and 15 two-digit hydrologic unit (HUC2) regions, tuned all models on data through 2020, and evaluate them on one common set of station/date/horizon keys at 1-, 3-, and 7-day leads, with a held-out 2021–2023 test window. The cohort is availability-selected rather than drawn, which bounds what any of it can generalize to; Section 3.6 and Section 6 state that bound rather than working around it.

Five questions organize the study. How much of a reported gain survives a strong reference? Which model class is actually most accurate on identical keys? How much of a reported spatial transfer survives whole-region holdout? In which issue-time-identifiable hydrologic states does a learned model still add skill? And — since every model above sees only issue-time information — how much predictability is withheld by that restriction rather than by the models, which we bound in Section 4.8 with realized future meteorology. The predictor whose skill we decompose, *ThermoRoute*, is described in Section 3.1; its architecture is the object under test rather than the contribution.

2 Data and design

2.1 The frozen panel and the cohort it defines

The development record is a single derived panel, `data_usgs/panel_usgs_120v2.parquet`, bound by a manifest (`data_usgs/frozen_panel_v1.json`) to a station registry (`data_usgs/station_registry_v1.csv`). Each of 120 sites carries one row for every calendar day from 2006-01-01 through 2020-12-31, giving 657,480

rows; missing observations are explicit nulls, not absent rows. The models consume seven raw variables: water temperature, streamflow, air temperature, precipitation, a relative-humidity proxy, Daymet daylight-period mean incoming shortwave radiation, and wind speed, from USGS NWIS daily values (United States Geological Survey, 1994), Daymet V4 (Thornton et al., 2022) and gridMET (Abatzoglou, 2013).

The cohort was not sampled at random from U.S. rivers, and nothing here treats it as though it were. Candidate selection rejected 1,345 of 1,465 candidate stations, mostly for lacking joint water-temperature and discharge availability. The retained cohort is availability-enriched, and the 2019–2020 interval participated in its construction, so it is development data in the strict sense, used for model tuning and diagnosis rather than as an independent test.

Observed water temperature is absent on about 15.8% of panel rows and discharge on about 2.8%. Among retained stations, 38 share a repeated hydrologic unit code with another retained station and 19 have another retained station within 10 km, so station-level independence is not tenable.

2.2 Temporal roles and the common key set

Temporal roles are fixed before any model is fitted, and a training sample is admitted only if its issue date and every one of its target dates fall inside the same partition.

Role	Dates	Rows	Observed WTEMP
Training	2006–2015	438,240	341,646
Validation	2016–2017	87,720	84,074
Calibration	2018	43,800	42,279
Development evaluation	2019–2020	87,720	85,621
Held-out test	2021–2023	—	observed

Models are tuned exclusively on data through 2020; the 2021–2023 window is held out and used only for the comparative evaluation of Section 4.4.

The key registry is a data rule, not a model outcome. A key is admissible when its issue date and target date fall inside the window, issue-date wa-

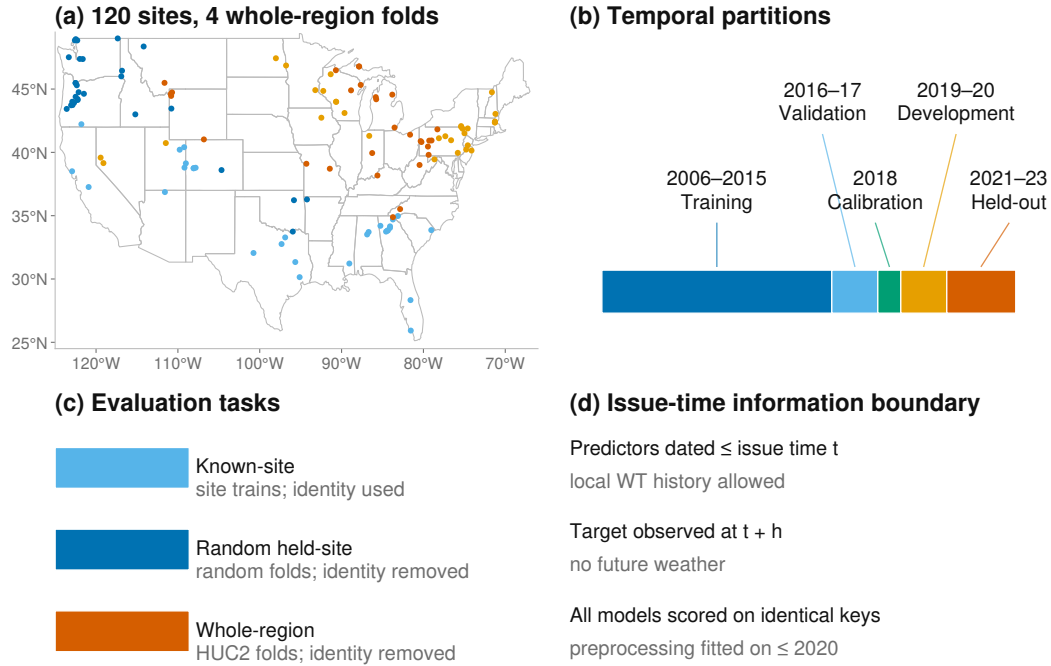


Figure 1. Study sites and evaluation design. (a) The 120 U.S. gauges colored by the four deterministic whole-HUC2-region folds. (b) Temporal partitions. (c) The three evaluation tasks: known-site forecasting, random held-site transfer, and whole-region gauged transfer. (d) Issue-time information boundary: all models are scored on the identical common forecast-key registry. No model-specific complete-case set is permitted.

ter temperature is finite and observed, a 32-day history window can be built, and target-date water temperature is finite and observed under the inclusive qualifier policy — evaluated independently at each lead. None of those conditions mentions a model. Every declared model is then required to produce a row on every admissible key, so a model cannot enter the comparison having declined the hard days: failing to predict is a failure, not an exclusion. This matters because the alternative construction — intersecting the days each model happened to score — is the second of the three inflation channels this study sets out to remove (Section 5.1), and a registry built that way would be indistinguishable from this one in a reported RMSE. On the development-evaluation partition the registry holds 249,072 station/date/horizon keys, 83,024 per lead; the held-out window carries its own registry of the same form. The 249,072 figure describes the development registry only and is never used for a held-out claim.

2.3 Evaluation-period inputs and the information boundary

The test interval is 2021-01-01 through 2023-12-31 for the same 120-site cohort. Meteorology is represented at each station coordinate and is not aggregated over upstream catchments, a real limitation for large basins (Section 6).

The primary information set uses provider values dated no later than each historical issue date and consumes no horizon-specific future weather forecast. This is a date-indexed retrospective hindcast, and not a re-execution of what a forecaster could have run on the day (Section 6): the as-issued provisional vintage of a gridded product cannot be reconstructed after the fact, so archiving requests, responses, timestamps, and checksums freezes the dataset actually evaluated without proving that identical values were available operationally at the time.

Daily mean water temperature (NWIS parameter 00010, °C), discharge (00060), and gage height (00065) come from USGS NWIS as the provider's 00003 daily-mean statistic on site-local dates. Discharge is served in $\text{ft}^3 \text{s}^{-1}$ and **is consumed in those units**: it enters the models only after per-station standardization, so the unit affects no reported quantity, but nothing in the pipeline converts it and a reader should not assume the panel column is in $\text{m}^3 \text{s}^{-1}$. The SI factors are $0.0283168 \text{ m}^3 \text{s}^{-1}$ per $\text{ft}^3 \text{s}^{-1}$ and 0.3048 m per ft . Gage height is retained as raw provenance and is not a model input. Every reported result is in °C.

Values are retained irrespective of approval qualifier, and the qualifier string is preserved rather than used to select sites, dates, or models. Two physical-range facts follow from that inclusive policy and are not corrected: 2,059 of 657,480 panel rows (0.31%) carry negative discharge, which at tidally influenced and regulated reaches is a real measurement rather than an error, and 1,304 rows carry water temperature between -0.9 and -0.1 °C. Observed water temperature spans -0.90 to 34.30 °C. The station-by-variable qualifier ledger, the sensor-continuity limitation, and the provider-specific day definitions that make this a date-indexed rather than an instantaneous comparison are in SI12. Test-window daily means are outcomes only; they are not read by model-selection, feature-selection, threshold-selection, calibration, or station-inclusion code, all of which is fixed on data through 2020.

A pooled-preprocessing sensitivity re-fits the model suite with pooled transforms in place of per-station ones; it is not a site-disjoint cohort (Section 3.4).

3 Methods

3.1 The predictor under test

ThermoRoute predicts at issue time t , station i , and lead h by adding a bounded learned residual to a damped-persistence anchor,

$$(1) A_{i,t+h} = c_{i,t+h} + \phi_i^h (y_{i,t} - c_{i,t})$$

where c is a seasonal climatology and ϕ_i a per-station decay fitted by no-intercept least squares on training-period consecutive-day anomaly pairs. The same object, fitted identically, is the damped-persistence reference.

The residual comes from a strictly left-looking temporal convolutional encoder over a 14-lag window, with a horizon-conditioned variable/lag router, a three-expert mixture, and a $\pm 1^\circ\text{C}$ algebraic bound on the deviation from the anchor. One model serves all three leads; 38,505 trainable parameters; five seeds averaged. The full specification is in Text S3. The architecture is the object under test, not the contribution: Section 4.3 shows an information-matched plain causal network reproduces it to within 0.023°C .

3.2 The reference set

The composition of the reference set is the most consequential methodological choice in this study, because it is what the reported gain is a gain over. Each member removes one alternative explanation for apparent skill.

Persistence ($\hat{y}_{-\{t+h\}} = y_{-t}$) quantifies the raw memory of the series. **Damped persistence**, equation (1) fitted on the training period, absorbs inertia and seasonality and is the reference for the primary comparisons. **Seasonal climatology** isolates the seasonal cycle alone. **LightGBM** (Ke et al., 2017) is the principal learned reference, run globally with site identity as a categorical feature and per station. A **global LSTM** with a station embedding represents the deep

Common anchor–residual formulation under matched information

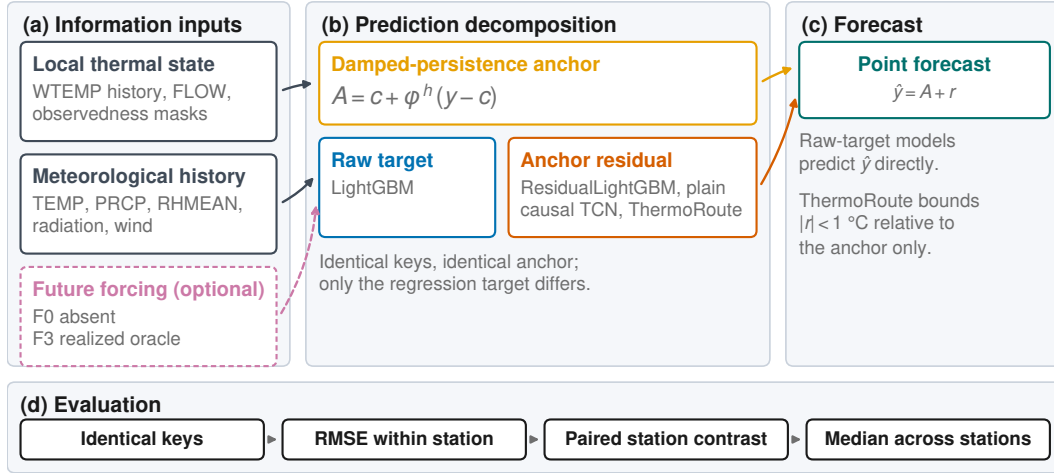


Figure 2. The formulation every compared model shares. LightGBM is a *raw-target* model with site identity; ResidualLightGBM, the plain causal TCN, and ThermoRoute predict a residual around the same anchor, with identical keys and information set. The figure is structural and carries no evidence from either period.

sequence family (Rahmani et al. (2021); Zwart, Diaz, et al. (2023)). A **plain causal temporal convolutional network** is the information-matched deep reference: it receives the same outcome-free issue-time inputs as the full model, is matched to within 2% of its parameter count (38,346 against 38,505), and predicts an unrestricted residual around the same anchor while omitting the relaxation proposal, router, mixture, and residual bound.

Tuning budgets are documented but *not* equalized across model classes: any statement about LightGBM here is scoped to this four-candidate procedure and this feature schema, not to gradient boosting in general. Seven one-factor architecture controls are deletion and intervention sensitivities and do not prove component necessity or identify a mechanism.

An air2stream-style hybrid reference (Toffolon and Piccolroaz, 2015) is fitted per station and scored on the same held-out keys as every other model (SI07). It is an unofficial empirical thermal-recurrence variant of the published formulation — not the official code and not a validated reproduction — so no process-side claim is attached to its scores.

260 3.3 Leakage control

261 Leakage control is a set of mechanisms checkable against the code and the arti-
 262 facts, not a statement of intent. Every predictor a primary model consumes carries
 263 a date no later than the issue date, and the encoder of Section 3.1 is strictly left-
 264 looking (Text S3). Standardization constants, the q90 event thresholds (2006–2015),
 265 the conformal offsets (2018), and the Platt calibrators (2018) are each fitted on data
 266 strictly preceding the interval on which they are applied. We record one honest gap:
 267 the issue-date auxiliary quantities behind the relaxation proposal and the regime
 268 gate carry no separate auxiliary-path mask, so the output is not guaranteed invari-
 269 ant to the fill values. A fresh interpreter replays every trained member and rejects
 270 merely self-consistent prediction files.

271 3.4 Two spatial partitions, separated from local adaptation

272 Random held-site splits are the usual way to report spatial generalization,
 273 but they can be optimistic: if a held-out gauge's neighbours remain in training, the
 274 model reaches that site's thermal regime through correlated forcing and shared pre-
 275 processing statistics. We therefore run a 2×2 factorial on the independent 2021–2023
 276 window separating the spatial partition from the local-adaptation policy (protocol
 277 v1; Section 4.5 and SI11).

278 The **random held-site warm-start** arm uses four balanced folds in which
 279 held-site histories still contribute to global panel preprocessing; it is the arm most
 280 comparable to published random-split results, and for that reason not the one we
 281 treat as the spatial test. The **held-region gauged transfer** arm packs the 15
 282 HUC2 groups into four folds of [30, 30, 31, 29] stations and holds out whole regions;
 283 the mean distance from a held-out station to its nearest training gauge is 289 km.
 284 *Local* adaptation lets a target station's own 2006–2015 history enter its climatology,
 285 damped anchor, and imputation medians; *pooled* adaptation fits those statistics over
 286 in-fold stations only.

287 Held-out sites still supply their own observed water temperature through the
 288 issue date. This is **gauged** transfer to a region whose gauges were not used in fit-
 289 ting; it is **not** prediction at a site with no observational record.

3.5 Conformal intervals

Split-conformalized quantile intervals are calibrated on the 2018 year only and are reported as an empirical marginal coverage diagnostic with width and interval score beside every coverage figure; no finite-sample or conditional-coverage claim is made (details and sensitivities in SI08).

3.6 Estimand, metrics, and comparison set

Two differently signed quantities are reported, and they are never combined. The paired effect is

(2) $\Delta\text{RMSE} = \text{RMSE}(\text{candidate}) - \text{RMSE}(\text{reference})$, in $^{\circ}\text{C}$, **negative** favors the candidate

and the skill score is

(3) $\text{skill} = 1 - \text{RMSE}(\text{candidate})/\text{RMSE}(\text{reference})$, dimensionless, **positive** favors the candidate

Equation (2) is the formal estimand and carries a $^{\circ}\text{C}$ unit; equation (3) is a bare signed number. Every value in Section 4 is labeled with which it is.

The sampling unit is the station. For each lead, unweighted RMSE is computed on the common daily keys per reportable station, and the primary effect is the unweighted median across stations of equation (2) with ThermoRoute as candidate; a station/lead cell is reportable only with at least 100 valid paired targets.

The comparison set is five head-to-head rows: ThermoRoute against damped persistence at 1 d, 3 d, and 7 d with a 0.00 $^{\circ}\text{C}$ reference margin, and against Light-GBM at 3 d and 7 d with a +0.05 $^{\circ}\text{C}$ numerical ceiling. That ceiling is a stated numerical reference, not a decision threshold, and carries no ecological or regulatory importance.

Two uncertainty procedures accompany each row, both clustered at HUC2: a one-sided p-value from exact enumeration of all 2^K whole-cluster sign vectors, and a 10,000-draw whole-HUC2 cluster bootstrap percentile interval for the median station effect, Holm-adjusted (Holm, 1979) over the five p-values. The cohort came

from an availability filter, not probability sampling of HUC2 regions, so exchange-
able region sampling is not established, and the clustered intervals and p-values
are approximate descriptive sensitivities, not decision evidence. No comparison is
written as a superiority, non-inferiority, equivalence, or parity statement, and none
generalizes to a national population.

4 Results

Three evidence grades are used, and they are not interchangeable.
Section 4.4 is *confirmatory*: its model suite, five-test family, estimand, margins,
and decision rules were fixed on data through 2020 and the 2021–2023 window was
opened only to score them. Sections 4.1–4.3 are *development diagnostics* on the
2019–2020 partition, which participated in cohort construction and informed model
selection, so they are the basis on which the suite was fixed rather than a test of it;
values there are five-seed ensemble means on the 249,072 common development keys.
Sections 4.5–4.10 are *post-outcome descriptive* analyses, specified after the held-out
window had been opened; nothing in them is a confirmatory test, and no amount of
subsequent care converts a post-outcome specification into one.

The descriptive results are nevertheless the largest effects this study measures,
and three of them are quoted in the Abstract, the Key Points, and the Conclusions.
That is a deliberate choice with a cost: a reader weighing them must know they
were not pre-specified. Every quotation of a Section 4.5–4.10 result therefore carries
the descriptive label at the point of use, and SI20 maps each headline number to its
grade, its artifact, and when its specification was written. They are hypotheses this
cohort generated, not findings it confirmed.

Every value in Section 4 is an unweighted station median over reportable sta-
tions on the relevant common key registry (`outputs/final/`), in °C. Pooled RMSE
appears only as a Supporting Information sensitivity and is never labelled a station
median.

4.1 The reference model, not the architecture, sets the reported gain

Most of a reported gain does not survive a strong reference. On identical keys, the same fixed model reports a seven-day median station skill of +0.251 against persistence and +0.038 against damped persistence — a factor of 6.6 between two numbers that differ only in the denominator of equation (3). A study reporting only the persistence column would describe this model as gaining a quarter of the seven-day error budget. A study reporting both columns describes the same fitted weights as gaining four percent.

Lead	Persistence	Damped persistence	LightGBM	LSTM	ThermoRoute
1 d	0.803	0.774	0.578	0.662	0.631
3 d	1.576	1.406	1.280	1.323	1.291
7 d	2.217	1.738	1.649	1.679	1.657

Development-period (2019–2020) station-median RMSE, °C.

The reference is itself a fitted object, and its construction moves the headline. We rebuilt the anchor under seven predeclared one-factor variants, rescored each against the *published* ThermoRoute predictions so that every difference is attributable to the reference alone. Seven-day skill against damped persistence then ranges from +0.025 to +0.063, with the published +0.038 in the middle (`outputs/final/anchor_sensitivity.parquet`); the low end comes from a decay fitted directly at each lead, a *stronger* reference. The seven-day headline is the least stable, so the number this paper quotes most often is the most sensitive to how the reference was specified: the better the anchor, the less the learned model adds.

4.2 The most accurate model on this panel is a gradient-boosted tree

It does, and we report it directly. On the development window the tree ensemble has the lowest station-median RMSE at all three leads: 0.578, 1.280, and 1.649 °C against ThermoRoute's 0.631, 1.291, and 1.657 °C. The paired station-level differences (`ThermoRoute` – `LightGBM`, equation 2) are +0.046 °C at 1 day, +0.008 °C

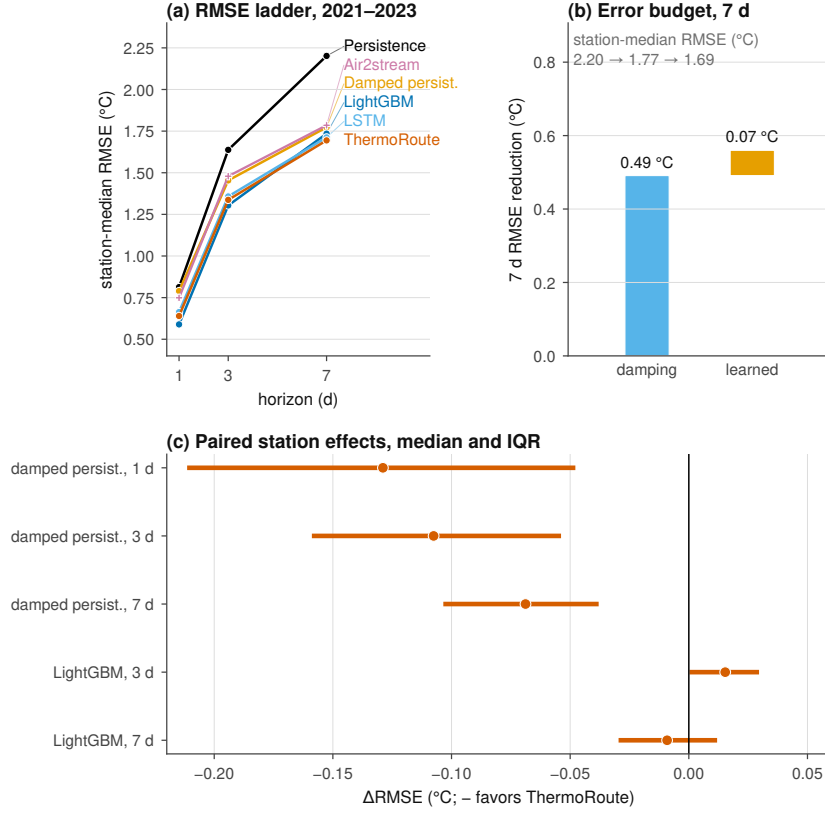


Figure 3. Decomposition of reported skill on the held-out 2021–2023 window (116 reportable stations). (a) Station-median RMSE by lead, every compared model. (b) The seven-day error budget: 0.49 °C of the reduction comes from damping the anchor and 0.07 °C from the learned residual. (c) Paired station effects for the five sealed tests, median and interquartile range; negative values favor ThermoRoute. Panel (c) shows the dispersion of the paired effects, not the cluster-bootstrap intervals of Table 2, which are narrower.

at 3 days, and -0.005 °C at 7 days, with ThermoRoute win rates of 0.00, 0.40, and 0.60 across the 120 stations. The one-day difference is outside the frozen five-test family; as an exploratory check it is $+0.035$ °C (sign-flip $p = 1.0$ under the one-sided family convention). The architecture's remaining argument is not accuracy but the ± 1 °C algebraic bound of Section 3.1, which held on 100.00% of audited rows — a property of the construction rather than a fitted outcome.

4.3 An information-matched plain convolutional network reproduces the architecture

Very little of the architecture's structure is doing work. Against the information-matched plain causal temporal convolutional network (same inputs,

same anchor, same optimiser budget, 38,346 against 38,505 parameters) the full architecture's paired median station effect, computed within seed on identical keys, ranges across the five seeds from -0.0003 to -0.0105 °C at 1 day and -0.0035 to -0.0226 °C at 7 days — every seed favors the full model. The one-factor deletions agree: removing the temporal encoder is the largest single effect (1-day station-median RMSE 0.631 to 0.679 °C), while removing the router, mixture, dynamic prior, fixed-relaxation constraint, or residual bound moves the 1-day figure by at most 0.004 °C (SI09). These are five-seed deletion and intervention sensitivities on paired keys; they do not prove component necessity.

4.4 Held-out 2021–2023 evaluation

The held-out evaluation applies the Section 3 model suite and comparison set, fixed on data through 2020, to 2021-01-01 through 2023-12-31. Every number is an unweighted station median over reportable stations on the common held-out key registry ($n = 116$; per-model metrics in SI07). Station-median skill is $+0.250$ against persistence and $+0.038$ against damped persistence, and the tree with site identity leads at one and three days (0.589 and 1.304 °C). At seven days ThermoRoute reaches 1.694 °C against the tree's 1.735 °C. Against the information-matched plain convolutional network the paired station median is -0.015 °C at seven days in ThermoRoute's favour, on the sign convention of equation (2); this contrast is outside the sealed family and carries no decision.

Each test is read at the margin it was sealed at. The five-test family is not five tests of the same shape. The three damped-persistence rows were sealed as superiority tests at a 0.00 °C margin; the two LightGBM rows were sealed as *non-inferiority* tests at a $+0.05$ °C margin, a strict numerical ceiling on allowable degradation fixed on 2026-07-22 under an attestation that no post-2020 outcome had been requested or inspected (protocols/route_a_confirmatory_v1.json, sealed by route_a_protocol_seal_v1.json). The margin is a number, not a judgement: the protocol's own rationale records that it is not derived from sensor precision, biological response, or regulation, and its wording limit forbids reading it as equivalence, parity, or ecological negligibility. All five rows satisfy their sealed decision rule — Holm-adjusted one-sided $p \leq 0.05$ and the whole percentile interval below the registered margin.

Two limits on that statement. Row 4 is a *supported non-inferiority* result at a lead where ThermoRoute is nevertheless the worse model: the effect is $+0.015$ °C with an interval excluding zero from above, so the tree wins at three days and wins by less than the ceiling. At seven days the paired median is -0.009 °C, in ThermoRoute's favour but with an interval covering zero, so that row establishes the ceiling and not a win. And the seal is repository-internal Git and SHA-256 evidence, which the protocol itself describes as sufficient for an honest owner and not proof against a repository owner rewriting history; there is no external timestamp and no independent custodian. A referee should read the confirmatory label as auditable rather than as attested.

Table 1 — every model on the common held-out keys. Station-median RMSE, MAE and bias in °C at each lead, 116 reportable stations, identical station/date/horizon keys for every row.

Model	RMSE 1	RMSE 3	RMSE 7	MAE 1	MAE 3	MAE 7	bias 1	bias 3	bias 7
Persistence	0.813	1.638	2.202	0.597	1.235	1.686	-0.000	-0.001	-0.004
DampedPersistence	0.789	1.454	1.773	0.591	1.100	1.340	-0.029	-0.076	-0.143
Climatology	1.899	1.902	1.903	1.485	1.486	1.485	-0.379	-0.369	-0.359
LightGBM	0.589	1.304	1.735	0.431	0.995	1.315	-0.025	-0.098	-0.196
LSTM	0.663	1.358	1.712	0.503	1.044	1.292	-0.001	-0.051	-0.137
PlainMLP-7var	0.690	1.374	1.720	0.520	1.054	1.311	-0.004	-0.059	-0.128
PlainCausalTCN-7var	0.646	1.330	1.710	0.477	1.021	1.299	0.000	-0.031	-0.122
Air2stream	0.719	1.459	1.825	0.559	1.115	1.366	-0.020	-0.110	-0.189
ThermoRoute	0.640	1.337	1.694	0.463	1.013	1.269	0.011	-0.029	-0.130

Table 2 — the five sealed tests, each at its registered margin. ThermoRoute is the candidate and 116 stations are reportable on every row. Station-level Δ RMSE in °C, negative favors ThermoRoute; whole-HUC2 cluster-bootstrap intervals; sign-flip p-values computed against the row's own margin and Holm-adjusted across the family (unadjusted $p \leq 6.1 \times 10^{-5}$ throughout — the floor for 15 clusters). Test identifiers, unadjusted p-values, and the decision components are

in `outputs/final/sealed_confirmatory_family_v1/`. The margin column is not decoration: rows 4 and 5 were registered as non-inferiority tests, and testing them against zero — as an earlier version of this table did — reports a registered hypothesis as a failure of a hypothesis nobody registered.

#	Reference	Lead	Margin	Δ RMSE	95% CI	Win	Holm p	Sealed decision
			(°C)	(°C)		rate		
1	DampedPersistence	1 d	+0.00	-0.129	[-0.199, -0.076]	0.90	1.5e-04	SUPPORTED
2	DampedPersistence	3 d	+0.00	-0.108	[-0.143, -0.074]	0.91	1.5e-04	SUPPORTED
3	DampedPersistence	7 d	+0.00	-0.069	[-0.084, -0.057]	0.95	1.8e-04	SUPPORTED
4	LightGBM	3 d	+0.05	0.015	[0.010, 0.025]	0.24	1.8e-04	SUPPORTED
5	LightGBM	7 d	+0.05	-0.009	[-0.017, 0.003]	0.59	1.8e-04	SUPPORTED

One-factor ablations neither help nor hurt systematically at any lead: fixing the dynamic prior and removing the bounded-residual constraint give station-median skill against damped persistence of +0.180 and +0.176 at one day against +0.173 for the full model (SI09). These are deletion and intervention sensitivities and do not prove component necessity; neither adds material point accuracy on the independent window.

4.5 Matched spatial-transfer experiment on 2021–2023

The development-period whole-region finding (SI19) is re-tested on the independent window with a matched 2×2 factorial design that separates spatial geometry from local adaptation. Station-agnostic gradient-boosted trees are fitted under four leave-HUC2-region folds and four balanced random folds over five split seeds, crossed with local and pooled preprocessing (protocol v1; SI11). The design moves the median nearest-training-gauge distance from 60 km under random held-site folds to 263 km under whole-region folds. Under local adaptation the region-minus-

random paired station penalty is $+0.006$ °C at one day, $+0.009$ °C at three days, and $+0.007$ °C at seven days for the raw-target tree.

This experiment does not resolve a geometry effect, and we do not report one. The penalty is the same size as its own resampling noise — the median per-site standard deviation across the five split seeds is 0.004 – 0.007 °C — and four whole-region folds provide four independent regional observations. We report the penalty as *below this design's resolution*.

The pooled arm is defective: its preprocessing reused fold 0's statistics across folds 1–3, whose held-out stations lie inside fold 0's training set (a recorded defect, DLOG-012). We report no adaptation main effect from it, and no number from it is carried into the Discussion or the Conclusions.

4.6 Hydrologic conditions governing incremental skill

The remaining learned gain concentrates in specific hydrologic states. Station thermal memory, the anomaly half-life of the train-fitted damped-persistence anchor — an anomaly half-life, not an e-folding time — has a station median of 6.9 days (interquartile range 5.0–11.9). At seven days the median memory gain is 0.49 °C and the median learned gain 0.07 °C (Figure 3b), and the median per-station fraction of the error reduction delivered by seasonal memory is 0.875. The station correlation between log half-life and learned gain is -0.40 at one day: longer-memory stations leave less for the learned model to add at short leads.

Issue-time-identifiable states. Thresholds are station-specific empirical quantiles fitted on the 2006–2015 training period only (SI11). At seven days the learned model's median paired ΔRMSE over damped persistence is -0.069 °C on all keys; it is largest under low thermal anomaly (-0.229 °C) and rapid recent cooling (-0.170 °C), and reaches -0.088 °C under rapid recent warming. No issue-time-identifiable state reverses the ordering.

The full sixteen-stratum table is in SI11 rather than here, on two grounds the numbers themselves supply. The 100-key reportability rule of Section 3.6 qualifies a station on its *all-keys* record; inside a stratum the same station is scored on far fewer, and the median per-station key count falls to 71, so those rows rest on

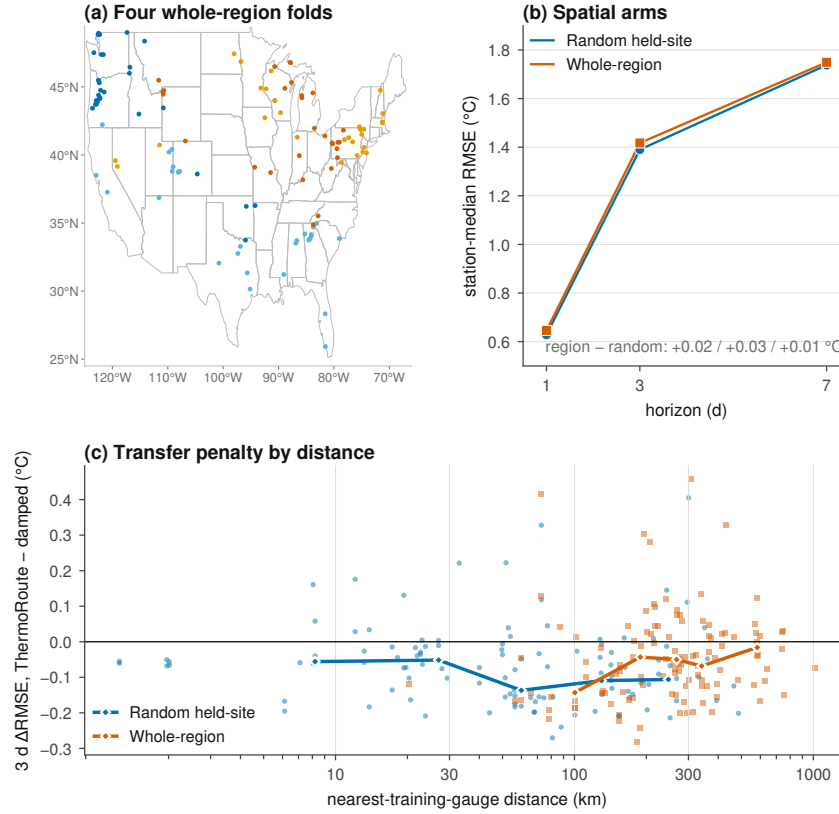


Figure 4. Spatial transfer under matched random-site and whole-region holdouts on the independent 2021–2023 window. (a) The four whole-HUC2 folds. Panels (b) and (c) show **ThermoRoute** under the earlier region-transfer run (`outputs/conventional/region_transfer_metrics_2021_2023.csv`), *not* the station-agnostic raw-target tree of the Section 4.5 factorial. That is why panel (b) annotates a region-minus-random gap of $+0.02/+0.03/+0.01$ °C where the text quotes $+0.006/+0.009/+0.007$ °C: two different models under two different fold constructions, not two estimates of one number. Both sit inside their own resampling noise and neither resolves a geometry effect, which is the only conclusion either supports. Panel detail and pooled-arm station-set caveats in SI11.

roughly a fifteenth of the evidence the all-keys row does. And the strata are not a multiplicity family: nothing was adjusted across them, no row carries an interval, and none was registered.

Retrospective outcome-conditioned diagnostics. Stratifying by the target outcome — retrospective, not an issue-time state — days that subsequently warm rapidly reach -0.30 °C at seven days, and days that subsequently cool rapidly -0.18 °C. On the coldest target decile the anchor alone is marginally better ($+0.03$ °C), the only stratum in which the learned correction does not pay for itself. Those

strata select the days on which a damped anchor must fail, and a manager cannot identify them at issue time; we read them as a diagnostic of where the anchor is weak rather than an operational benefit.

4.7 What local thermal state is worth

Section 4.5 could not resolve a spatial effect because every arm kept the held station's own thermal history. This section removes that history instead of the geography, which is the contrast the design can actually resolve.

Four nested information levels are fitted under whole-region holdout: **L0** keeps everything; **L1** pools the climatology and damped rate over training stations; **L2** removes every target-site water-temperature input, leaving discharge and meteorology; **L2-U2** removes discharge as well. Prohibited inputs are dropped from the design matrix rather than filled; all 24 proofs return a maximum absolute difference of exactly zero, and a negative control re-admitting a single water-temperature lag is caught with a 147 °C shift (`outputs/final/information_ladder_v6_authority_v1/`).

Local thermal state is the dominant information in this problem.

Withholding it costs 1.3–1.7 °C, at every lead, at every one of 116 stations. That is two to thirteen times the value of realized future meteorology (Section 4.8) and roughly seventy times the architecture effect of Section 4.3. A study that keeps the target gauge's recent readings and then reports a model comparison is comparing models inside the regime where the largest available information is already present.

Almost all of that value is the recent sequence, not the station's long history. Pooling a station's climatology and damped rate costs 0.01–0.16 °C; removing its recent readings costs 1.17–1.71 °C. The two ablations are not the same size and should not be described with the same word: pooling a station's long-run statistics is nearly free, and withholding its recent observations is not. Neither is a test of prediction at a site that has never been instrumented, which this cohort cannot supply, because a station with no thermal record also has no target to score against (Section 6). Once water temperature is gone, also removing discharge changes station-median RMSE by 0.009–0.084 °C with every interval covering zero.

Spatial geometry matters, but only once local history is gone. Section 4.5 measured a geometry penalty of 0.004–0.009 °C and could not resolve it. Running the same four levels under random-site holdout shows why: the penalty is 0.006–0.014 °C at L0 and 0.073–0.129 °C at L2. The station-level double difference is +0.066 to +0.124 °C, and every whole-HUC2 interval excludes zero. A random-site split in a dense network measures something closer to interpolation than transfer, but the cost of that substitution is invisible while the held gauge still supplies its own thermal history.

4.8 What future meteorology would be worth

Every result above is issue-time-only, so none of it says whether the small residual learned gain reflects a limit of the models or a limit of the information they were given. This section refits the tree models with the *realized* meteorology over $t+1 \dots t+h$ substituted for the issue-time forcing, on the same cohort and the same 116 reportable stations. F3 is a **retrospective realized gridded meteorological oracle**: an upper bound on what perfect weather information would be worth, not a forecast product and not an operational gain. It is also *post-outcome* — the 2021–2023 window was already open when it was specified — so it is a descriptive reanalysis and not a confirmatory test. It is quoted in the Abstract and the Key Points, under that label, on the terms set out at the head of Section 4.

Table 3 — value of realized future meteorology. Station-first forcing value $V_F = \text{median}_i[\text{RMSE}_i(F0) - \text{RMSE}_i(F3)]$ in °C. CI is a 10,000-draw whole-HUC2 cluster bootstrap; MDE is the smallest effect this cohort's cluster structure can resolve at $\alpha = 0.05$.

Model	Lead	RMSE F0	RMSE F3	V_F	95% CI	MDE
LightGBM	1 d	0.614	0.451	0.130	[0.081, 0.189]	0.031
LightGBM	3 d	1.330	0.765	0.542	[0.415, 0.721]	0.152
LightGBM	7 d	1.742	1.063	0.627	[0.423, 0.803]	0.172
ResidualLightGBM	1 d	0.606	0.446	0.125	[0.084, 0.183]	0.028
ResidualLightGBM	3 d	1.327	0.752	0.578	[0.406, 0.704]	0.160
ResidualLightGBM	7 d	1.722	1.087	0.605	[0.445, 0.795]	0.151

Every value is three to four times its own minimum detectable effect, where the architecture effects of Section 4.3 are smaller than theirs, and no leave-one-HUC2-out range changes sign. Formed as a station-level double difference, the forcing value rises by $+0.415^\circ\text{C}$ [0.301, 0.541] from one to three days but by only $+0.074^\circ\text{C}$ [0.046, 0.101] from three to seven (LightGBM). Future weather buys most of what it can buy by day three; the seasonal and per-year strata are in `outputs/final/forcing_regime_v5_observed_inference_authority_v1/`.

The gain is event-scale weather information, not seasonal phase. A placebo whose design and decision rules were sealed before the outcome existed de-ranges the meteorology vector across dates *within each station-month*, preserving each station's climate and seasonal phase and destroying only the correspondence between a forecast key and its weather. It retains 0.0% of the true value at one day, 2.0% at three and 8.8% at seven, and the sealed rule P1 is satisfied at every model and lead. A second sealed control displaces the realized future by a whole week, leaving local weather persistence intact and removing only the exact dates; the +7-day arm retains -0.1% , 0.7% and 14.0% , and P4 is satisfied.

What the forcing value is worth as a warning. Both arms were scored on warm-tail exceedance and signed rapid change, with thresholds fitted per station on the 2006–2015 training period only. At seven days with the raw-target tree, the probability of detection for exceedance of a station's training 95th percentile rises from 0.18 to 0.55 while the false-alarm ratio *falls*, from 0.36 to 0.20, so this is not a base-rate trade. These are deterministic point predictors and none emits a calibrated

event probability; the comparison is an oracle bound in event space exactly as it is in RMSE space.

Which rivers depend on future weather. Of six attributes declared before the relationships were inspected, the strongest is the one the physics predicts: the anomaly half-life of the station's own damped anchor, $\rho = -0.49$ $[-0.66, -0.16]$ at seven days. Rivers that hold a thermal anomaly for longer carry their own state forward and need tomorrow's weather less. One attribute is a nuisance control and it is not null — stations with more complete training records show a smaller forcing value ($\rho = -0.29$ $[-0.45, -0.07]$) — so part of the apparent physical gradient may be a record-quality gradient. We report these as exploratory covariation on a fixed, non-probability cohort with about nine effective spatial clusters, not as an attribution, and we do not fit a multivariable model the design cannot carry.

The forcing value is future air temperature. At seven days, air temperature alone recovers 0.595 °C of the 0.627 °C full value, while withholding it and giving everything else its realized future retains only 0.129 °C.

About half the oracle survives a real forecast. The F2a arm reproduces that temperature-only arm with an archived forecast substituted for realized air temperature at prediction time, and recovers 49–62% of it across leads and both trees. The series is a **fixed-lead composite**, not a coherent trajectory from one initialization and not an as-issued operational archive; the archive spans 2021–2023, so this is a measurement of those years; and the arm supplies air temperature only. The F2b archived-vintage arm remains unrun under the same seal.

4.9 Future weather is worth less, not more, when local observations are withheld

Read separately, Sections 4.7 and 4.8 invite a reading neither tested: that better forcing could recover what the withheld observations cost. We crossed the axes, refitting the F3 forcing arm at L0 and at L2 under the same whole-region folds and level-legal anchor as Section 4.7. The contrast is between two input sets at the same

stations, and it is stated that way throughout; it is not a comparison between an instrumented and an uninstrumented river.

Table 4 — value of realized future meteorology by information level.
Station-first paired values in °C under whole-region holdout, 116 reportable stations, 10,000-draw whole-HUC2 cluster bootstrap.

Forcing value	1 d	3 d	7 d
At L0, gauged	0.116 [0.059, 0.174]	0.474 [0.299, 0.693]	0.580 [0.366, 0.779]
At L2, obs. withheld	0.038 [0.020, 0.072]	0.331 [0.249, 0.496]	0.596 [0.431, 0.804]
Interaction, L2 – L0	−0.076 [−0.098, −0.056]	−0.147 [−0.193, −0.079]	+0.016 [−0.035, +0.099]

Future weather is worth *less* to a model that has lost the local gauge, not more: the interaction is negative at one and three days and excludes zero at both. Realized meteorology earns its value by correcting a trajectory, and at short leads that trajectory is the persistence anchor built from the station's own recent readings; without it the model falls back on a pooled climatology too coarse for tomorrow's weather to sharpen. The two information sources are complements at the leads where local state dominates.

This analysis is post-outcome and descriptive, and is quoted in the Key Points under that label. It is the study's most interpretable descriptive result and its least tested one — the kind of finding that ought to be pre-registered before it is believed (outputs/final/forcing_information_interaction.v1/).

4.10 The estimator matters only where the information does not

Every contrast to this point varies information and holds the model class fixed. This one does the reverse, and it is the contrast on which the paper's central claim can fail. A plain causal temporal convolutional network is fitted in each cell of the crossing, information-matched to the trees, and compared with the residual-target tree that shares its target formulation. Positive means the network is worse

(full crossing, both holdout geometries and per-cell minimum detectable effects in
outputs/final/architecture_geometry_interaction_v1/).

In the regime this literature reports, the two implementations differ by about a hundredth of a degree. At L0 under issue-time information the residual-target tree and the information-matched network differ by at most 0.009 °C at any lead, against station-median errors of 0.53–1.74 °C, and the same holds under random-site holdout, the geometry this literature uses.

That sentence is about two fitted objects, not two model families. One tree and one plain causal network do not span the space of estimators, their tuning budgets were documented rather than equalized (Section 3.2), and a graph, recurrent, or differentiable-hybrid model is not represented at all. The contrast supports *the estimator not being where this problem's difficulty lives*, on this panel, under matched information. It does not support a general claim that model class is worth 0.01 °C in river-temperature prediction.

Where information is scarce, the estimator starts to matter. Remove the local gauge and the tree wins by 0.02–0.19 °C. The architecture-by-information interaction is positive in all six geometry-by-lead cells and resolved in three of them — at one day under both holdouts and at three days under random-site — while the two seven-day cells and whole-region at three days have intervals covering zero. What makes the estimator matter is the missing local information, not the spatial extrapolation the whole-region design bundles with it.

That result depends on the network having been trained far enough, and its own lineage said it had not: under the shared 40-epoch budget the median best epoch at L2 was 37, with a third of cells still improving at the cap, while L0 converged well inside it. A budget that binds only where the claim lives is the information-mismatch failure relocated to the optimiser, so the L2 cells were refitted with a 300-epoch cap and patience 20. The median best epoch moves to 119 and nothing reaches the new cap, and the penalty does not shrink — it grows, to 0.249, 0.152 and 0.065 °C, with every change interval covering zero. Longer training makes the network worse against the held region while its in-fold validation still improves, which is what an information-scarce regime looks like from the optimiser's side.

The reported figures are the shorter-budget ones, so they are the conservative pair (outputs/final/tcn_convergence_sensitivity_v1/).

Given perfect future weather, the sequence model is the better one.

At L0 with F3 the sign reverses, to -0.096 $[-0.177, -0.038]$ at seven days. That is the paper's one resolvable architecture gain, and it requires an oracle: an argument for sequence models *conditional on good forcing*, not for the regime the literature benchmarks. The L0 triple difference sits below each cell's minimum detectable effect (0.006 , 0.012 and 0.018 °C), and so does the same contrast at L2. Both are therefore underpowered, and neither is evidence of absence: an effect below the smallest one a design can detect is an effect that design cannot speak about, and the sealed decision contract says so — failure to reject is not equivalence. Section 4.4's two LightGBM rows are the only negligibility statements this paper owns, because they carry the only margin fixed before the outcome was seen. No margin is declared for any architecture contrast, and one chosen now would be chosen knowing the answer.

5 Discussion

5.1 Why these numbers are smaller than published gains

Our reported gains are far smaller than those commonly published for daily river temperature, and the difference is one of design, not model quality. Three choices account for it.

The first is the reference. Where a study quotes skill against persistence or climatology, the number it reports is comparable to our $+0.203$ to $+0.251$ column, not to our $+0.038$ to $+0.168$ column; on a variable with this much day-to-day memory the two differ by up to a factor of 6.6 for the same fitted weights. We suggest damped persistence as the minimum reference for this variable.

The second is the spatial partition, and here our design was not able to measure what it set out to measure. Holding out whole regions moves the median distance to the nearest training gauge from 60 to 263 km, but every arm retains the held station's thermal history, and the measured geometry penalty (0.004 – 0.009 °C) sits inside its own resampling noise (Section 4.5). We make no claim about the size

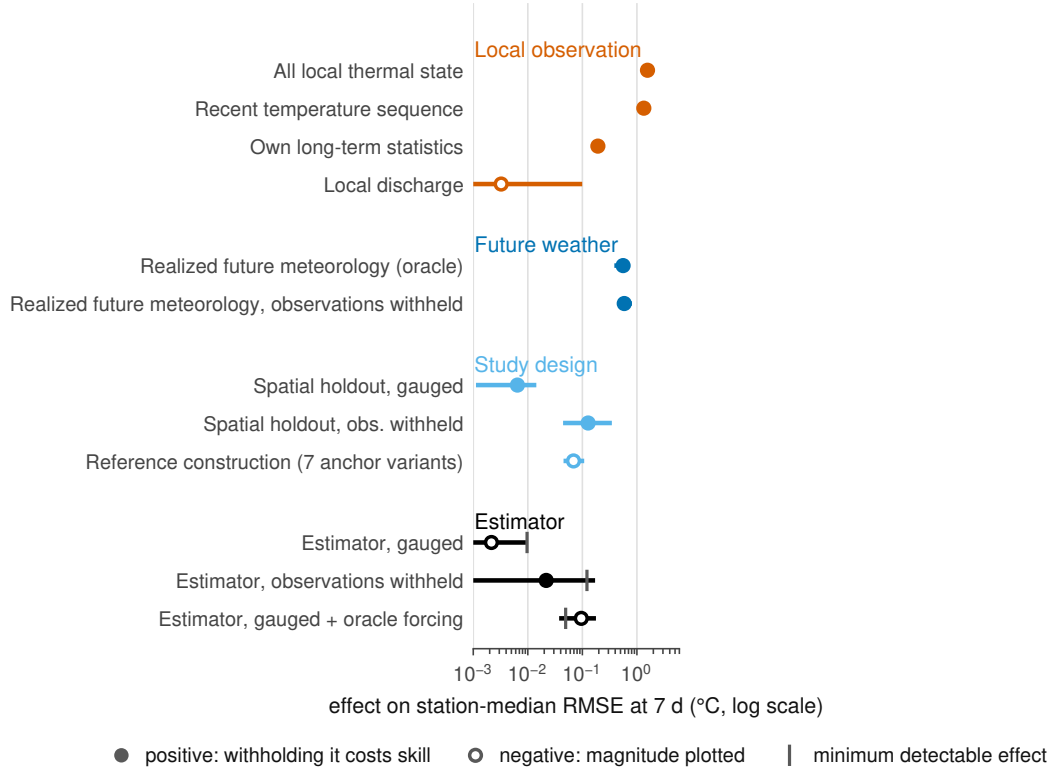


Figure 5. What each axis of the study is worth, at seven days. Station-first paired effects in °C on a log axis, damped-residual tree throughout; open markers are negative effects plotted at their magnitude, and the tick marks the minimum detectable effect where one is defined. Local observation, future weather, study design and the estimator span three orders of magnitude. The axes are separate conditional designs and are compared, never summed. Every row is post-outcome and descriptive except the last, and every bar is a whole-HUC2 cluster-bootstrap interval except the last. *Reference construction* is the range of the seven-day effect across the seven predeclared one-factor anchor variants of Section 4.1 — a spread over specifications, not a sampling interval — and it is on this axis because every other row is conditional on one anchor while the anchor is itself a fitted object. It is the row that says how much of the rest of the figure is a property of the reference rather than of the river.

of the spatial effect on this cohort, in either direction. A random-site split in a dense network measures something closer to interpolation than to transfer.

The third is the key set and the preprocessing boundary. Scoring each model on its own complete-case set, and fitting scaling, imputation, climatology, or interval offsets on windows that include the evaluated interval, move error downward without leaving a visible trace. We do not assert that the published literature is affected;

removing both channels here left the constrained deep model behind a gradient-boosted tree at the 1- and 3-day leads and matched at 7 days.

5.2 What a benchmark of this geometry can carry

The evaluation design of this study is stronger than its inferential reach. Clustered inference over a gauge panel rests on resampling or sign-flipping whole spatial units, and its accuracy depends on having enough units, of comparable size, that the asymptotics they invoke are not a fiction. Fifteen HUC2 groups with an inverse-Herfindahl effective count of 9.54 and a largest group holding 21.7% of stations is not enough, and the sampling design was never exchangeable across regions. We therefore present the clustered intervals and p-values as approximate descriptive sensitivities and do not attach decision weight to them.

The usual practice would report a station-level interval or a paired test across sites, and nothing in such a paper would tell a reader that the effective number of clusters is under ten; the difference is disclosure, not the data and not the model.

5.3 What transfers

This study is narrow by construction: a common-key, clustered comparative benchmark of point predictors at gauged sites, not an attempt to replace process-based thermal models, river-network graph models, or differentiable hybrid formulations (Jia et al. (2021); Rahmani et al. (2023); Zwart, Diaz, et al. (2023)). Every arm consumes the target site's observed water temperature through the issue date (Section 3.4), so the paper is silent on prediction at locations with no thermal record.

What is portable is not the architecture but three controls: an identical key registry across models, calibration and preprocessing statistics fit strictly backwards in time, and spatial partitions by whole hydrologic region rather than by site. On this cohort the reference choice moved a reported number by more than the differences between architectures did; whether that ordering holds across the published literature is a question this study motivates rather than settles.

6 Limitations

Scope. The design is a descriptive benchmark on a fixed cohort: comparisons are station-first and approximate (15 HUC2 clusters; effective cluster count 9.54; largest share 21.7%; non-probability sampling), and no interval, p-value, or ranking claim is decision evidence. Section 4.7 withholds a station's own thermal record from the model, which is not the same as predicting at a location that has never been instrumented: the cohort, the climatology and the evaluation keys all still come from gauged sites. Genuinely ungauged prediction is not quantified here, and no operational-replay claim is made. Throughout, "causal" describes time ordering only, never causal inference.

Cohort, measurement, and design. The cohort trades a modest discharge channel (removal costs $+0.042$ °C at 1 day; SI19) against wide spatial coverage; 950 of 1,465 candidates were excluded for missing joint flow. Point-scale meteorology is used at station coordinates rather than basin integrals. The air2stream-style hybrid is an unofficial empirical variant of the published formulation (Toffolon and Piccolroaz, 2015) — not the official code, not a validated reproduction — so no process-side claim is attached to its scores.

Reference construction. The damped anchor is a fitted object and the seven-day headline moves by a factor of 2.5 across defensible constructions of it (Section 4.1). Every number stated against damped persistence is therefore conditional on the anchor specified in Section 3.1, which is primary because it was fixed before the held-out window was opened, not because it is the strongest available.

Threshold and archive scope. The ± 1 °C algebraic bound, event-threshold quantiles, and reportability thresholds (100 paired keys) are development-selected; the q90 event threshold is a statistical tail diagnostic with no biological, ecological, or regulatory interpretation. Development-period spatial analyses and robustness probes are archived in SI19.

7 Conclusions

We evaluated daily river water-temperature prediction at 120 stable U.S. gauges across 15 hydrologic regions, scoring every model on one common set of sta-

tion/date/horizon keys at 1-, 3-, and 7-day leads, with all statistics fitted strictly backwards in time and a held-out 2021–2023 test window.

Three findings survive. First, the reference model governs the reported gain: the deep predictor's seven-day station-median skill is +0.250 against persistence but +0.038 against damped persistence. Second, model ranking does not favor architectural constraint: a gradient-boosted tree with site identity has the lowest station-median RMSE at 1- and 3-day leads (0.589 and 1.304 °C). Both LightGBM rows satisfy their sealed +0.05 °C non-inferiority bound, which is a statement about a pre-declared numerical ceiling and not about equivalence, parity, or ecological negligibility; at three days the tree is the better model by +0.015 °C while staying under that ceiling. Third, the spatial-partition question the design was built to ask is unresolvable in the regime it was posed in and resolvable just outside it. With the held station's own thermal history retained, holding out whole regions rather than random sites changes station-median RMSE by less than the five-seed resampling spread. Withhold that history and the same contrast separates (Section 4.7), which is why the geometry of a split cannot be assessed independently of how much local information the model keeps.

Every model here is issue-time-only. A retrospective analysis (Section 4.8) bounds what perfect future meteorology could add at 0.13 °C at one day and 0.54–0.63 °C at three and seven days. A placebo sealed before the outcome existed removes 91–100% of that value: the gain is event-scale weather information, not seasonal phase.

A published skill score is not interpretable without its reference model, its spatial partition, and its information set. The comparisons are descriptive for this fixed, availability-selected cohort: clustered intervals and p-values are approximate sensitivities, not decision evidence.

Open Research Statement

Data availability. The derived daily panel (`panel.usgs_120v2.parquet`), the stable station registry (`station_registry_v1.csv`), their manifests, and the test-window acquisition record are deposited as one versioned dataset at [DATA DOI TO BE MINTED] under [DATA LICENCE TO BE ASSIGNED]. Neither can be assigned

before the byte-level rights review of Section 6 completes, because a derived product does not inherit the most permissive of its upstream terms; the deposit is described here as planned and specified, not as existing.

Primary observational sources. Water temperature and discharge come from USGS NWIS (United States Geological Survey, 1994), meteorology from Daymet V4 R1 (Thornton et al., 2022), and wind speed from gridMET (Abatzoglou, 2013), each under its own terms.

Material that cannot currently be redistributed. Where redistribution terms are unresolved, the deposit carries the request specification and a SHA-256 manifest in place of the bytes; exclusions are enumerated in SI16.

Software availability. The analysis software, the per-stage manifests, and the Python 3.12 dependency lock with package hashes are archived at [SOFTWARE DOI TO BE MINTED]. The MIT licence on the source code covers neither the observational data nor the archived provider responses.

Reproduction. Sections 4.1–4.3 reproduce from the archived panel, bundles and pinned environment; Section 4.4 from the archived test-window acquisition record. Bit-level equality is expected for the tree models and agreement to a documented tolerance for the neural members; tolerances and digests are in the Supporting Information. Reproduction has not yet been carried out by an operator independent of the authors on an independent host, and this statement is not a claim that it has.

Acknowledgments

[FUNDING TO BE COMPLETED — each funder with its award number, or an explicit statement that the work received no external funding.]

[COMPUTATIONAL RESOURCES TO BE COMPLETED — the facility or facilities on which the model suite was fitted.]

We acknowledge the U.S. Geological Survey for the National Water Information System, the ORNL Distributed Active Archive Center for Daymet V4 R1, and the Northwest Knowledge Network for gridMET. Acknowledgment of a data provider is not an endorsement of this analysis by that provider.

Competing interests. [COMPETING INTERESTS TO BE COMPLETED
— a declaration is required from every author, including the explicit statement that
none exists if that is the case.]

Author contributions. [CREDIT ROLES TO BE COMPLETED — assign
each author to the applicable CRediT roles: conceptualization, methodology, soft-
ware, validation, formal analysis, investigation, resources, data curation, writing —
original draft, writing — review and editing, visualization, supervision, project ad-
ministration, funding acquisition. The signed intake form is docs/FAIR_SUBMISSION_
READINESS_AND_TEMPLATES.md §2.]

Supporting Information

Supporting Information contains the cohort and registry description with
the candidate-rejection ledger (SI01); the issue-time information boundary and
the product-compatibility bridge (SI02); model equations and unit conventions
(SI03); the analysis protocol, redesign chronology and decision log (SI04); the com-
parison set and its fields (SI05–SI06); all-model scores on the exact common keys,
including the fitted air2stream-style hybrid reference (SI07); probability and relia-
bility schemas with the measured degenerate-interval table (SI08); architecture and
information-matched controls with seed and budget provenance (SI09); the tempo-
ral coverage audit (SI10); the matched spatial factorial and the cluster geometry at
HUC2 through HUC8 (SI11); outcome quality control and qualifier evidence (SI12);
the history-dependent external arm (SI13); missingness and failure cases (SI14);
reproduction hashes, commands and environment parity fields (SI15); the rights
and data dictionary (SI16); the development-period spatial analyses and robustness
probes relocated from the main text (SI19); and the method, estimands and seal-
ing of the local-information, forcing, geometry and architecture arms of Sections
4.7–4.10 (SI20).

Figures S1–S3 describe the cohort geometry, the temporal roles and issue-time
boundary, and the model and calibration dataflow; Figures S4–S8 expand Section
4.4 and the hydrologic-state mechanism of Section 4.6; Figure S9 is a development-
period conformal-calibration sensitivity, labelled as such in the figure, and must not
be compared numerically with any held-out-window figure.

References

- Abatzoglou, J. T. (2013). Development of gridded surface meteorological data for ecological applications and modelling. *International Journal of Climatology*, 33, 121–131. (gridMET / METDATA) doi: 10.1002/joc.3413
- Arsenault, R., Brissette, F., & Martel, J.-L. (2018). The hazards of split-sample validation in hydrological model calibration. *Journal of Hydrology*, 566, 346–362. doi: 10.1016/j.jhydrol.2018.09.027
- Bakar Siddik, M. A., Turner, S., Gomez-Velez, J., Tanim, A., & Kao, S.-C. (2026). Advancing stream temperature prediction with a generalizable large-sample framework across CONUS river reaches. *Journal of Hydrology*, 666, 134780. doi: 10.1016/j.jhydrol.2025.134780
- Barclay, J. R., Topp, S. N., Koenig, L. E., Sleckman, M. J., & Appling, A. P. (2023). Train, inform, borrow, or combine? approaches to process-guided deep learning for groundwater-influenced stream temperature prediction. *Water Resources Research*, 59, e2023WR035327. doi: 10.1029/2023WR035327
- Caissie, D. (2006). The thermal regime of rivers: a review. *Freshwater Biology*, 51, 1389–1406. doi: 10.1111/j.1365-2427.2006.01597.x
- Chang, S. Y., Schwenk, J., & Solander, K. C. (2025). Deep learning advances Arctic river water temperature predictions. *Water Resources Research*, 61(6), e2024WR039053. doi: 10.1029/2024WR039053
- Corona, C. R., & Hogue, T. S. (2025). Machine learning in stream and river water temperature modeling: a review and metrics for evaluation. *Hydrology and Earth System Sciences*, 29, 2521–2549. doi: 10.5194/hess-29-2521-2025
- Feigl, M., Lebedzinski, K., Herrnegger, M., & Schulz, K. (2021). Machine-learning methods for stream water temperature prediction. *Hydrology and Earth System Sciences*, 25, 2951–2977. doi: 10.5194/hess-25-2951-2021
- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70. Retrieved from <https://www.jstor.org/stable/4615733>
- Jia, X., Zwart, J., Sadler, J., Appling, A., Oliver, S., Markstrom, S., ... Kumar, V. (2021). Physics-guided recurrent graph model for predicting flow and temperature in river networks. In *Proceedings of the 2021 siam international conference on data mining (sdm)* (pp. 612–620). SIAM. (Preprint: arXiv:2009.12575)

- doi: 10.1137/1.9781611976700.69
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., ... Liu, T.-Y. (2017). LightGBM: a highly efficient gradient boosting decision tree. In *Advances in neural information processing systems 30 (neurips 2017)* (pp. 3146–3154). Retrieved from <https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html>
- Kleměš, V. (1986). Operational testing of hydrological simulation models. *Hydrological Sciences Journal*, 31(1), 13–24. doi: 10.1080/02626668609491024
- Kratzert, F., Klotz, D., Herrnegger, M., Sampson, A. K., Hochreiter, S., & Nearing, G. S. (2019). Toward improved predictions in ungauged basins: Exploiting the power of machine learning. *Water Resources Research*, 55(12), 11344–11354. doi: 10.1029/2019WR026065
- Liu, J., Shen, C., O'Donncha, F., Song, Y., Zhi, W., Beck, H. E., ... Lawson, K. (2025). From RNNs to Transformers: benchmarking deep learning architectures for hydrologic prediction. *Hydrology and Earth System Sciences*, 29, 6811–6828. doi: 10.5194/hess-29-6811-2025
- Mohseni, O., Stefan, H. G., & Erickson, T. R. (1998). A nonlinear regression model for weekly stream temperatures. *Water Resources Research*, 34, 2685–2692. doi: 10.1029/98WR01877
- Nearing, G. S., Kratzert, F., Sampson, A. K., Pelissier, C. S., Klotz, D., Frame, J. M., ... Gupta, H. V. (2021). What role does hydrological science play in the age of machine learning? *Water Resources Research*, 57(3), e2020WR028091. doi: 10.1029/2020WR028091
- Padrón, R. S., Zappa, M., Bernhard, L., & Bogner, K. (2025). Extended-range forecasting of stream water temperature with deep-learning models. *Hydrology and Earth System Sciences*, 29(6), 1685–1702. doi: 10.5194/hess-29-1685-2025
- Philippus, D., Corona, C. R., & Hogue, T. S. (2026). Daily stream water temperature modeling and forecasting for ungaged watersheds at the CONUS scale. *Journal of Hydrology*, 675, 135620. doi: 10.1016/j.jhydrol.2026.135620
- Rahmani, F., Appling, A., Feng, D., Lawson, K., & Shen, C. (2023). Identifying structural priors in a hybrid differentiable model for stream water temperature modeling. *Water Resources Research*, 59, e2023WR034420. doi: 10.1029/2023WR034420

- 936 Rahmani, F., Lawson, K., Ouyang, W., Appling, A., Oliver, S., & Shen, C. (2021).
 937 Exploring the exceptional performance of a deep learning stream temperature
 938 model and the value of streamflow data. *Environmental Research Letters*, *16*,
 939 024025. doi: 10.1088/1748-9326/abd501
- 940 Sadler, J. M., Appling, A. P., Read, J. S., Oliver, S. K., Jia, X., Zwart, J. A., &
 941 Kumar, V. (2022). Multi-task deep learning of daily streamflow and wa-
 942 ter temperature. *Water Resources Research*, *58*, e2021WR030138. doi:
 943 10.1029/2021WR030138
- 944 Seibert, J., Vis, M. J. P., Lewis, E., & van Meerveld, H. J. (2018). Upper and
 945 lower benchmarks in hydrological modelling. *Hydrological Processes*, *32*(8),
 946 1120–1125. doi: 10.1002/hyp.11476
- 947 Thornton, M. M., Shrestha, R., Wei, Y., Thornton, P. E., & Kao, S.-C. (2022).
 948 *Daymet: daily surface weather data on a 1-km grid for North America, version*
 949 *4 R1*. ORNL DAAC, Oak Ridge, Tennessee, USA. doi: 10.3334/ORNLDAA/2129
 950
- 951 Toffolon, M., & Piccolroaz, S. (2015). A hybrid model for river water temperature as
 952 a function of air temperature and discharge. *Environmental Research Letters*,
 953 *10*, 114011. doi: 10.1088/1748-9326/10/11/114011
- 954 Topp, S. N., Barclay, J., Diaz, J., Sun, A. Y., Jia, X., Lu, D., ... Appling, A. P.
 955 (2023). Stream temperature prediction in a shifting environment: explaining
 956 the influence of deep learning architecture. *Water Resources Research*, *59*,
 957 e2022WR033880. doi: 10.1029/2022WR033880
- 958 United States Geological Survey. (1994). *USGS water data for the nation*. Na-
 959 tional Water Information System (NWIS) daily values accessed via the datare-
 960 trieval Python client; parameter codes 00010, 00060, 00065 with statistic
 961 code 00003. (Accessed 2026; <https://waterdata.usgs.gov/nwis>) doi:
 962 10.5066/F7P55KJN
- 963 Zwart, J. A., Diaz, J., Hamshaw, S., Oliver, S., Ross, J. C., Sleckman, M., ...
 964 White, E. (2023). Evaluating deep learning architecture and data assim-
 965 ilation for improving water temperature forecasts at unmonitored locations.
 966 *Frontiers in Water*, *5*, 1184992. doi: 10.3389/frwa.2023.1184992
- 967 Zwart, J. A., Oliver, S. K., Watkins, W. D., Sadler, J. M., Appling, A. P., Corson-
 968 Dosch, H. R., ... Read, J. S. (2023). Near-term forecasts of stream temper-

969 ature using deep learning and data assimilation in support of management
970 decisions. *JAWRA Journal of the American Water Resources Association*,
971 59(2), 317–337. doi: 10.1111/1752-1688.13093